GLS University
Faculty Of Computer Applications & Information Technology
iMScIT/iMCA Semester IX

Capstone Project - 2026-27

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Company/Organization Name:

Company Contact No:

Company Email Id:

Project Title: HeartSense AI: An AI-Powered Heart Disease Prediction

Frontend Technologies : React.js

Backend Technologies: Django

List of Hardware / Sensor / Processor (if Project on Embedded System) :

Project Description:

HeartSense AI: An AI-Powered Heart Disease Prediction and Clinical Decision Support System is an intelligent healthcare application that uses Artificial Intelligence (AI) and Machine Learning (ML) to predict the risk of heart disease based on patient medical data. The system analyzes key health parameters such as age, blood pressure, cholesterol, ECG results, chest pain type, and heart rate to generate accurate risk predictions. It employs machine learning algorithms like Random Forest, Logistic Regression, SVM, and XGBoost, along with Explainable AI (SHAP/LIME) to provide transparent and interpretable results. The application also offers personalized health recommendations and displays prediction outcomes through an interactive dashboard. Designed as a clinical decision support tool, the system helps healthcare professionals and patients make informed decisions while promoting early diagnosis and preventive cardiovascular care.

Project Modules:

1. User Authentication Module
   - User registration and login
   - Secure password encryption
   - Role-based access (Patient, Doctor, Admin)

2. Patient Data Management Module
   - Enter patient medical details
   - Update and delete patient records
   - View patient history
   - Store records securely in the database

3. Data Preprocessing Module
- Handle missing values
- Remove duplicate records
- Normalize and encode data
- Feature selection and scaling

4. AI/ML Prediction Module
- Train machine learning models
- Predict heart disease risk
- Compare multiple algorithms
- Generate prediction confidence score

5. Explainable AI (XAI) Module
- Display feature importance
- Explain prediction using SHAP or LIME
- Show factors contributing to the risk level

6. Risk Assessment Module
- Classify patients into:
  - Low Risk
  - Moderate Risk
  - High Risk
- Display overall risk score

7. Personalized Recommendation Module
- Suggest lifestyle improvements
- Recommend diet and exercise plans
- Provide preventive healthcare advice
- Encourage medical consultation when necessary

8. AI Chatbot Module (Optional)
- Answer basic heart health questions
- Explain prediction results
- Provide educational information
- Guide users through the application

9. Dashboard & Visualization Module
- Display prediction results
- Show patient statistics
- Visualize health parameters using charts
- Present model performance metrics

10. Report Generation Module
- Generate patient health reports
- Export reports as PDF
- Print prediction summaries
- Maintain historical reports

11. Admin Module
    - Manage users and patient records
    - Monitor system usage
    - Manage datasets
    - Retrain or update AI models
    - View analytics and system logs

12. Database Management Module
    - Store user information
    - Store patient medical records
    - Save prediction history
    - Backup and restore data